



More Than One Fix

Quick facilitation notes & answer key

Goal

Students find a loop bug that has TWO valid fixes — change the repeat count or change the blocks inside — show both reach the goal, then choose one and justify it (C3.2, with C3.1 'writing' on Ex 3–4).

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block app or Python Turtle projected; the printed worksheet + a pencil. Write 'total actions = repeat × blocks inside' on the board so students can adjust either number.

Run it (about 30 min)

1. Model the worked case: repeat 4 { clap, clap } = 8, but the goal is 4.
2. Show BOTH fixes — halve the repeat (repeat 2) OR keep one clap inside (repeat 4 × 1). Each = 4.
3. Students do Exercises 1–4 (Ex 4 is the stretch: choose + justify), solo or in pairs.
4. Share: land that a bug can have more than one right fix, and a coder chooses the clearest.

Answer key (checked by the run-gate)

Ex 1 — Fix A: repeat 3 ($3 \times 2 = 6$). Fix B: one clap inside ($6 \times 1 = 6$).

Ex 2 — Fix A: repeat 5 ($5 \times 2 = 10$). Fix B: one hop inside ($10 \times 1 = 10$). Fix B changes fewer blocks (delete one hop).

Ex 3 — Fix A: repeat 4 { stomp, stomp } = 8. Fix B: repeat 8 { stomp } = 8.

Ex 4 (challenge) — Fix A: repeat 4 { clap, clap, clap } = 12. Fix B: repeat 6 { clap, clap } = 12. Any reasoned choice accepted (e.g. 'Fix B changes only one number').

Watch for & differentiate

Common slip: stopping at the first fix and missing the second.

Support: write 'total = repeat × blocks inside' and let students change either number to hit the goal.

Extend: ask for a goal where only one kind of fix works (an odd number of actions can't be reached by halving a two-block loop).

Success looks like

The student finds both a repeat-count fix and a blocks-inside fix that reach the goal, then chooses one and gives a one-sentence reason (Ex 4).